Everything from Two

Rise of Form, chapter draft v0.6, Project secret3, Row B, August 11, 2026

A note before the chapters. This is the interpretation row of the program: a reading offered to everyone, in harmony with a spiritual view of the world and in honest tension with the assumption that matter comes first. Nothing in these pages is a proof of the reading. Wherever a claim is mathematical, it is not merely quotable but checkable: it points to the working papers of the program, where a compiler has verified it. The reading is an invitation. The equations are load-bearing. They are never the same thing, and this book will never pretend they are.

This chapter does three things. First it builds: a working computer, piece by piece, out of the two moves alone. Then it proves the mountain: everything computable, from anything, is one and the same everything, and the two moves stand on its summit. And then it asks the one question the mountain cannot answer, and follows it, honestly, to a door. One word will carry the whole way, so fix it now. Form is anything that can be written down, copied, and compared. A mark, a shape, a symbol, a pattern, a structure: in this chapter these are all form, and the word never means more than that.

The Workshop

In the late 1970s, in a hallway at MIT, a group of students led by Danny Hillis built a computer out of Tinkertoys. Wooden spools, wooden sticks, fishing line, about ten thousand parts. When you made your move on its front panel, the whole scaffold of wood clattered through its motions and answered. It played tic-tac-toe perfectly. It never lost. It sits in a museum now, and it makes a point no lecture makes as well: computation does not care what it is made of. Silicon, brass gears, wooden spools, falling marbles, it is all the same computer. Computing is not a substance. It is a pattern of moves. Which raises the workshop question this section exists to answer. If it can be made of anything, how little can it be made of? The opening chapter announced the answer, two moves, and told the story of the man who found it. This section rolls up its sleeves and builds.

Start with the oldest component in the catalogue: a decision. Every computer, wooden or otherwise, must at bottom be able to choose. This or that. Watch how little that takes. Let true be K itself. Offered two things, true keeps the first:

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K x y  ->  x        one move
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And let false be S K, two symbols. Offered two things, it takes the second, in two moves, one share, one drop:

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S K x y  ->  K y (x y)  ->  y
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Both behaviors are checked in the proof module, kTrue and skFalse. Now notice what we did not have to build. There is no if machine, no switch, no gate. To decide between a and b, you simply apply the decision to them. Hand true the pair and it hands back a. Hand false the pair and it hands back b. Choice is not a component in this workshop. Choice is what application already was.

Next component: storage. A computer must hold two things together and give them back on demand. Here is a box that does it, built from seven symbols: for any two contents x and y, take S (S I (K x)) (K y), where I is the crown's three symbols from the opening chapter. This term simply waits. Hand it any key f, and in six checked moves, the theorem is pairBeta, it presents both contents to

the key: f x y. Hand it true, and true keeps the first: the box yields x. Seven moves in all, pairFst. Hand it false, and false takes the second: the box yields y. Eight moves, pairSnd. The proof of that last fact is worth a sentence, because it shows how this workshop composes. In the module, pairSnd is literally two earlier theorems glued end to end: the box opening, then false doing its work. Build once, reuse forever. The workshop's parts are theorems, and they snap together like the Tinkertoys.

And a computer must count. The opening chapter already stocked this shelf: a number, in the calculus, is an iterator, do it n times, with the successor built from the Answer section's composition. The arithmetic of the working papers runs on exactly this stock. So the shelves hold decisions, boxes, and numbers. Machinery enough for one honest demonstration: a complete program, run before the reader's eyes, every step one of the two rules and nothing else. Three sentences first, on how to read what follows. Writing two things side by side means: hand the right one to the left one, so F x means hand x to F. The leftmost thing that can act, acts. And parentheses only group, so (K y) F is the parcel K y being handed the key F. That is the whole notation. The program is the humblest in the catalogue, take the second thing out of the box. Here is its entire life, with F standing for the false key, S K:

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(box x y) F                    the box meets the key
(S I (K x)) F  ((K y) F)       share: the key is given to both halves
(S I (K x)) F  y               drop: the parcel K y, handed F, opens to y
F  ((K x) F)  y                the crown returns F (two moves, inside)
F  x  y                        drop: the parcel K x, handed F, opens to x
K y (x y)                      share: the false key starts its work
y                              drop: and keeps the second thing
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Eight moves in all, four shares and four drops, two of them folded inside the crown's line, and the box has handed over its second treasure. That is a computer program, complete, run to the end, with nothing under it. No processor, no electricity, no gates. Only the two rules of the opening chapter, applied eight times. Everything your phone did in the last second breaks down, in principle, into runs exactly like this one, unimaginably many of them. Not one step of any of them is anything but a share or a drop. The Tinkertoy machine showed that the material does not matter. The workshop shows something sharper: the pattern itself needs only two verbs. And notice, before the view widens, what the eight moves never once required: that any of it mean anything. The box did not know it was a box. The key did not know it was false. The run would have gone the same way if we had called the symbols pebbles, or angels, or nothing at all. Hold that thought. This chapter is on its way to it.

Notes and sources. The Tinkertoy tic-tac-toe computer was built by Danny Hillis, Brian Silverman, and friends at MIT in the late 1970s and is preserved in the Computer History Museum's collection. The encodings of truth, falsehood, and pairing are the classical Church encodings in combinator dress. The compiled anchors of this section, in the program's proof module: kTrue and skFalse for the two decisions; pairBeta, pairFst, and pairSnd for the box, the last composed from the two theorems named in the text; the crown skIdentity; and, for the arithmetic shelf, rLenSum and boxLen in the working papers. The module is total; the papers carry the verification.

One Mountain

In the spring of 1936, in Princeton, Alonzo Church published a definition of what it means for something to be computable. He built it from his calculus of substitutions. Within weeks, an ocean away, the young Alan Turing finished a paper defining the same idea with an imaginary machine, a read head crawling along a marked tape. Then he learned, with the sinking feeling every researcher knows, that Church had crossed the line first. That autumn Emil Post, in New York, working alone

and unaware of both, arrived by a third route. And Goedel, in lectures two years earlier, had carved a fourth out of pure arithmetic. Four pictures of doable, and they could hardly have looked less alike: a calculus, a machine, a worker with boxes, a tower of equations. Then the theorems came, and every picture drew exactly the same line. Not similar lines. The same class of tasks, task for task. Every definition proposed in the ninety years since has landed on it too.

What should we make of that? When four teams with different maps, different tools, and no knowledge of one another all reach the same peak, you stop believing the peak is something mapmakers invented. Computability behaves like something found. The reader of the opening chapter has stood in this spot before: many traditions, one triad. Here: many formalisms, one computability. And the two moves of this book stand on that peak as full citizens, because Schoenfinkel's calculus computes exactly what the machines and the substitutions compute. The two moves are the shortest name the mountain has.

Two more facts finish the map. First, the mountain has a cliff, not a slope. Below a certain line, systems are trivial. Above it, they are all exactly equal. Conway's Game of Life, a checkerboard pastime, is above the line and computes everything a supercomputer can. So is a one-line cellular rule. So is a folder of string-rewriting recipes. And there is no higher peak behind: nothing above the line can do more than anything else above it, ever. The line sits absurdly low, which is why the world is full of accidental computers. Second, the peak had an early flag on it. In January 1697, Leibniz sent his patron, Duke Rudolph August of Brunswick, the design for a medal celebrating his binary arithmetic. Around the rim ran a motto: omnibus ex nihilo ducendis sufficit unum. For deriving everything from nothing, one suffices. Leibniz read his arithmetic as an image of creation. That reading was his, and it is reported here as his; the medal, as far as we know, was never struck. Restruck for this book, the motto needs one edit: for deriving everything, two suffice.

Now the chapter can ask its real question. Look closely at the word everything. Everything here means: everything that can be done with marks. A symbol is a shape. A term is shapes arranged. A computation is shapes being rearranged by rules. The mountain, the whole of it, is a mountain of marks. The question of the rest of this chapter is simple to ask and very hard to sit with. Is meaning anywhere on this mountain?

Notes and sources. Church and Turing, both 1936, with Turing's appendix acknowledging Church; Post, 1936; Goedel's recursive functions from the 1934 Princeton lectures; the equivalences are classical. The universality of Life is in Berlekamp, Conway, and Guy, Winning Ways (1982); Rule 110 is Cook (2004). Leibniz's letter with the medal design is dated January 2, 1697. Combinatory completeness is Schoenfinkel (1924), carried into a science by Curry. The fuller tour of the summit, self-reproduction included, belongs to a later chapter of this part.

The Empty Symbol

A child asks what a word means. You reach for the dictionary, and the dictionary does the only thing a dictionary can do: it hands you more words. Some of those the child does not know either. So you look them up, and they are defined in words, which are defined in words. Chase every trail and you find what the compilers knew all along. The book is a circle. Every word is defined by the other words, and the whole heavy volume never once touches anything that is not a word. Cognitive scientists call this the dictionary-go-round, and it is not a flaw in dictionaries. It is an honest confession about form. A symbol can be traded for other symbols. It can be translated, transformed, rewritten. And at the end of every such trade, however long, what you hold is a symbol, exactly as empty as the one you started with. Swapping one form for another does not create meaning. It passes the question along.

The mountain makes this old observation exact, and that is its deepest service. Computation is the complete science of trading form for form. Complete: by the 1936 theorems, whatever can be done to marks as marks is on the mountain. All of it. So follow the thought one step. Suppose some clever shuffling of symbols produced meaning. That shuffling would be a way of doing things with marks, so it would be a computation, a citizen of the mountain like any other. Now ask what any computation hands you at the end. Marks. That is all it has to give: form in, form along the way, form out. The mountain cannot produce meaning for the same reason the dictionary cannot. Its every product is more of what it started with. And up close the reason shows. This book's own proof module is the exhibit. A checker verified every theorem in these chapters, thousands of steps without one slip, and it understood none of them. Nothing in it knows what a number is, or a move, or a book. And the module holds a two-line theorem that states the blindness formally. It is called substSteps, and it says: take any computation and rename every label in it, every named symbol riding in a term, to anything you please, and the computation runs step for step the same, to the same relabeled result. One note for exactness: S and K themselves are not labels but the machine, so the honest slogan is that the machinery is blind to the labels it moves. Said shorter, as this book will keep saying it: the moves never look at the names. They cannot. There is nothing in a move to look with.

The philosopher John Searle staged the same point as theater in 1980. A man sits in a room with a rulebook. Chinese questions come in through a slot. He matches shapes, follows rules, and passes perfect Chinese answers out, and he understands no Chinese at all. Does anything in that room understand Chinese? Forty years of debate have sharpened the question without dissolving it. The strongest reply says: maybe the man does not understand, but the room as a whole, man plus rulebook plus paper, does. Take that reply seriously; many serious people hold it. Then look at what the whole room is. Marks, and rules about marks. The question has not been answered. It has been moved. This is what the dictionary was telling us. The problem even has a name now, given by Stevan Harnad: the symbol grounding problem. How can symbols be about anything, when everything a symbol can ever do, by the completeness of the mountain, is become more symbols?

Now say the conclusion carefully, because it is easy to claim too much, and the point does not need it. This is not a claim that computers are dumb. Not a claim that the brain does nothing computable. Not a claim about what machines will or will not achieve. The claim is narrower and harder. Meaning is not something that shuffling form produces. Form-shuffling is a closed country, provably, and meaning is not in it. Wherever meaning comes from, it does not come from there. And there, says the medal, is everywhere that doing-things-to-marks can reach.

Notes and sources. The dictionary-go-round and the symbol grounding problem are Harnad (1990). The Chinese Room is Searle (1980); the systems reply is presented as the strong objection it is. The compiled emblem of the section is substSteps in the program's proof module, two lines; its companion exhibits are absCorrect and the checker's own working, which verified this book's formal spine while understanding none of it. The closure claim is the completeness theorem of the previous section.

The One-Second Test

There is a demonstration psychology teachers have run since 1935, and it never fails. Cards are printed with the names of colors, each name inked in the wrong color. The word red printed in green. The word blue in orange. The class is asked to go down the card naming only the ink and ignoring the words. The room slows to a crawl. People stammer, laugh, start over. The words will not stop being words. The effect is named after Stroop, it has been measured for ninety years, and one card shows it in ten seconds: a reader cannot, by any act of will, see writing as mere marks. The meaning arrives before permission. It cannot be declined.

Now widen the card to the world, and run the test this section is named for. Try to pass one second, a single second, without assigning meaning to anything. The ceiling is shelter. The sound in the next room is your child. The tightness in your chest is worry. The marks on this page are, unstoppably, a voice saying something to you. You cannot do it. No one has ever done it. Waking life is one unbroken act of taking form as something, and the act is so constant that we notice it the way fish notice water: only in a book like this, for a paragraph, before it closes over us again.

Put the two halves of this chapter side by side, and a border appears on its own. On one side, the mountain: everything form can do with form, complete, closed, and provably blind, the moves never looking at the names. On the other side, one second of anyone's life: meaning assigned constantly, involuntarily, everywhere. The most universal machinery we know never performs the act. The act never stops being performed. Something happens in every second of your life that is not a citizen of the mountain.

What kind of border is this? Be precise here, because it is the border of the sciences themselves, and it deserves better than a shrug or a sermon. Science works by a rule it chose on purpose. It admits what is public, repeatable, measurable. That is: form. The rule is the whole secret of science's power. It is also a fence. Even physics, the most exact science, tells us, in Russell's and Eddington's old observation, the pattern of the world, and stays silent about what the pattern is a pattern of. So when meaning-as-lived fails to appear in science's inventory of the world, that is not a discovery that meaning is absent. It is the method meeting its own fence, a fence it built, at the edge of form. Mistaking that fence for the edge of the world is the one error this chapter exists to mark.

The standard reply deserves its paragraph, because it is the working creed of half the field, and it is serious. Meaning, it says, is just use. A word's meaning is its role in the traffic of a life. The brain's aboutness is a very complicated causal wiring. Describe every function, and there is no residue left over; the itch you feel is a habit of grammar. Weigh that honestly. Then look at what any description of use, role, and wiring is made of. Form. More of it, better organized, still on the mountain, still never looking at the names. And the question walks back in through the door it was shown out of: used by whom, and useful to whom? This book will not pretend to settle here what philosophy has not settled in a century. It does what a Part One is allowed to do: it locates. The search for meaning is not one project among a life's projects. The one-second test shows it is the act underneath all of them, the thing every reader of every page is doing right now. And the complete science of form, by its own completeness theorem, has told us where not to look. That could sound like a bleak place to stop. It is not. The border has a gate, and the gate is closer than anything else in the world. The last section of this chapter walks up to it.

Notes and sources. The interference of reading with color naming is Stroop (1935), replicated continuously since. The structural limit of physical description is Russell, The Analysis of Matter (1927), and Eddington, The Nature of the Physical World (1928), the same Eddington who named the arrow of time in the opening chapter; the position is known today as structural realism and is mainstream. The meaning-as-use tradition begins with Wittgenstein's Philosophical Investigations; the functionalist reply is held, in different keys, by Dennett and much of cognitive science; the text claims location, not victory. The one-second test asks only the reader's own second. No compiled anchor is claimed for this section beyond those of the last, because its subject is exactly what the module cannot contain, and saying so is the point.

The Inside

Stand in a hardware store, in front of the wall of paint chips. Pick two neighbors from the same row. You can see the difference between them. Now try to say it. Warmer? Both are warm. Deeper? Not quite. The difference is right there, plain to your eye, and there is no word for it, and there never will be, because the numbers do not work out: people reliably tell apart millions of shades, and a lan-

guage holds a few dozen color words. Between what you can see and what you can say lies a gulf that is not poetry. It is measured. And psychology has known it from its first book. William James, founding the field in 1890, described consciousness as a stream and insisted that its most alive parts slip through every net of words: we ought to be able to speak of a feeling of and, a feeling of if, a feeling of but, he wrote, as readily as a feeling of blue or cold. We cannot. The inner life was always richer than any form we could give it, and the founder of psychology said so on page one, as it were, of the science.

Modern psychology went further, with a beeper. In one long-running method, people carry a pager that goes off at random moments, and each time it sounds they write down whatever was in their inner life in the last instant before the beep. Trained across thousands of samples, the reports keep containing something the researchers had to invent a name for: unsymbolized thinking. A thought that is definite, present, and clearly there, and that is in no words, no pictures, no symbols of any kind. Not a feeling of vagueness; a precise thought with no form. For some people it fills a quarter of their sampled moments. Add the quieter clinical fact that people who cannot describe their feelings at all still measurably have them, and the picture is consistent from every angle psychology has tried: inner life is not made of the forms we dress it in. The forms come later, and they always come up short.

Philosophy sharpened this into its most famous thought experiment. Mary is a scientist who knows everything physical there is to know about color: every wavelength, every retinal cell, every pathway in the brain, every fact that can be written down. She has learned it all inside a black-and-white room. One day she steps out and sees a red apple. Question: does she learn anything new? Almost everyone's gut says yes. She knew all the form there was, all of it, and the seeing was still not in the knowing. Philosophers call what she meets that morning by a careful phrase, coined by Thomas Nagel: what it is like. There is something it is like to see red, and no amount of structure adds up to it. One distinction makes this precise, and it is worth one sentence. What a mind can report, compare, and use, its traffic of information, is form, and it lives happily on the mountain of this chapter. The experience itself, the what-it-is-like, is the other thing. Honesty requires both footnotes, and they make the point stronger, not weaker. Some philosophers reply that Mary gains only a new ability, not a new fact, the way you can know all about bicycles and still learn to ride one. And Jackson, who invented Mary, later changed his own mind about what she proves. Yet notice what every reply keeps: nobody, on any side, denies that the seeing was different from the knowing. The datum survives its own debate. At the very least, and this modest version is enough for us, there is a gap that no description of form has closed.

Why is the gap where it is? Because it was built in, on purpose, at the founding. In 1623 Galileo drew the line that made physics possible. Keep, he said, what can be counted and measured, shapes, sizes, motions, the form; and set aside how things taste, sound, and look to someone, the qualities, as no business of the new science. It was a stroke of genius and it worked for four hundred years. But it means the silence of science about experience is not a discovery science made. It is the founding subtraction, echoing. And no scientist saw this more clearly than a man this book has already leaned on twice: Arthur Eddington, the astronomer who proved Einstein right in 1919, who named the arrow of time in the opening chapter, and whose observation about physics built this chapter's fence. He gave the fence its best parable, in a later book: a scientist trawls the sea with a net of two-inch mesh, examines a lifetime of catches, and announces a law of nature: all fish are larger than two inches. The net decides in advance what can be caught. Physics, Eddington said, is such a net, woven of measurement, and what it catches is structure, form, pattern, and only that.

And already in 1928, in the very book that named the arrow of time, he had taken the step this section has been walking toward. If our nets catch only pattern, is the inner nature of the world simply unknowable? No, he said. There is exactly one place in the universe where anyone knows what

something is from the inside, instead of catching its pattern from without. That place is your own consciousness. You do not infer your inner life from readings. You are it. It is the one sample of the world's stuffing that anyone holds, and it is known by acquaintance, not by form.

So the chapter can end where the author of the fence himself ended, at the gate. The proposal, scoped exactly: there is at least one level of existence that is not form. It is not exotic and it is not far away. It is the inside of this very second, the only thing in the universe you know otherwise than through pattern, the thing you do not observe but are. Everything else, stars, stones, other people, reaches you as form through the nets. This one thing does not reach you at all; it is where the reaching arrives. Serious philosophers hold this position today under sober names, and pressed hard it opens onto large questions, whether the inside is everywhere, whether it is one or many, what the old traditions meant by it, and this book defers every one of them, by name, to its later parts, with the contract in hand and the skeptic in her chair. The reader who suspects the inside will someday be shown to be more form is in serious company too, and this book will keep that company honest. Here, at the end of the chapter that proved the mountain and priced the symbol, only the modest claim is made, and it is enough: the fence is real, and so is the gate, and you have never once stood anywhere else. The Tinkertoy machine earns its last appearance. It never lost. It never once knew it was playing. You know. Hold that difference. It is the gate in the fence, and the rest of this book walks toward it.

Notes and sources. The discrimination of shades against the poverty of color vocabulary is the standard psychophysics, discussed as nuance ineffability in Raffman, Language, Music, and Mind (1993). James's feelings of and, if, and but are from The Principles of Psychology (1890), the Stream of Thought chapter. Unsymbolized thinking is documented in the descriptive experience sampling program of Hurlburt and colleagues (Hurlburt and Akter, 2008), method stated in the text; the clinical fact is the alexithymia literature (Sifneos, 1973). Mary is Jackson (1982, 1986); the ability reply is Lewis and Nemirow; Jackson's later change of mind is on record (2003); the phrase what it is like is Nagel (1974); the report-versus-experience distinction is Block (1995); the modest gap is Levine (1983). Galileo's subtraction is in The Assayer (1623), told for our time in Goff, Galileo's Error (2019). Eddington's net parable is from The Philosophy of Physical Science (1939); his acquaintance step, one place known from inside, is from The Nature of the Physical World (1928), the book that also named the arrow; Russell's structural point of the same years is The Analysis of Matter (1927), and the position's living descendants are the structural realists and Russellian monists (Strawson, 2006; Goff, 2019), reported, not adopted. The module has nothing to check in this section, and that is the section's point, made once more.